

RECOVERY GAIN IS OPERATING-POINT DEPENDENT IN PRUNED TEXT-TO-AUDIO DIFFUSION

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ABSTRACT

Recovery fine-tuning is usually evaluated at one inference setting, even though the gain it adds may depend on how a compressed generator is used. We study this effect in released AudioLDM-M checkpoints at 65% and 83% pruning, using paired generations across duration and prompt domain. At 83% pruning, CLAP recovery rises from +0.085 at 3.84 s to +0.244 at 10.24 s. A symmetric 20,000-step intervention shows that changing the fine-tuning duration from 3.84 to 10.24 s does not shift the gain toward the longer training duration, with $\Delta J = -0.035$ [-0.064, -0.004]. Beyond 10.24 s, recovery does not clearly increase. Recovery transfers strongly to held-out Clotho captions but remains an order of magnitude smaller on hip-hop, where dense anchors remain well above chance. These results establish recovery gain as operating-point dependent and motivate reporting paired recovery across displaced operating points rather than a single post-fine-tuning score.

Index Terms— text-to-audio generation, diffusion models, structured pruning, recovery fine-tuning, evaluation

1. INTRODUCTION

Structured pruning can make diffusion models cheaper by removing complete channels or blocks from the denoising network. The price is an immediate loss in model capacity, so modern compression pipelines rarely stop at pruning. They add a recovery stage that fine-tunes the smaller network and then judge the final checkpoint against its dense counterpart. This pattern appears in image diffusion [1–3] and has recently been applied to AudioLDM [5].

The usual evaluation answers an important systems question. It tells us whether the compressed checkpoint is competitive under a particular benchmark recipe. It does not tell us how much of that result comes from recovery, or whether the same recovery persists when the generation conditions change. Those questions are different because the final score of a recovered model mixes the state of the pruned checkpoint with whatever fine-tuning adds under the tested conditions.

We therefore treat recovery as a paired effect. For a pruned checkpoint P and its fine-tuned counterpart P+FT, the recovery gain is the change from P to P+FT under the same prompt, noise and inference setting. We call that setting the inference *operating point*. Requested duration is one axis, prompt domain is another, and sampler configuration provides a third. This framing turns a statement such as “the model recovered” into a question that can be tested across the conditions in which the model may actually be used.

The distinction is increasingly relevant as text-to-audio evaluation moves beyond single aggregate scores. CLAP remains a standard automatic proxy for prompt alignment [6–8], while recent

benchmarks place more emphasis on robustness and generalization [20]. Compression studies have not yet adopted this perspective systematically. Recovery is still commonly summarized by the post-fine-tuning score at one benchmark operating point.

We study the released pruned and recovered AudioLDM-M-Full checkpoints of Singh et al. [5]. The released 10^6 -step recovery exhibits a large duration interaction and selective transfer across prompt domains. We then intervene on the most natural explanation. Two new checkpoints are fine-tuned from the same pruned model for the same 20,000 steps, one at 3.84 s and one at 10.24 s. If the favorable evaluation duration were determined by the duration used during adaptation, the 10.24 s-trained checkpoint should show the stronger long-minus-short recovery interaction. It does not.

The resulting contribution is an evaluation result with a falsifiable boundary. Recovery gain varies strongly with the operating point, yet its duration dependence does not follow the fine-tuning duration at the matched 20,000-step budget. Across domain, recovery remains strong on Clotho but much smaller on hip-hop even though the latter battery discriminates the dense model from chance. Together these results motivate evaluating the recovery increment itself, rather than assuming that one post-recovery benchmark score describes the checkpoint more broadly. Expanded results and executable provenance are retained in the companion repository [21].

2. BACKGROUND AND MOTIVATION

2.1. Recovery is part of the compression pipeline

Diffusion compression methods increasingly optimize for the model that exists *after* adaptation. Diff-Pruning combines structured removal with retraining [1]. BK-SDM removes residual and attention blocks and transfers behavior through distillation [2]. TinyFusion explicitly selects architectures according to their expected performance after fine-tuning [3]. In each case, the useful compressed system is defined jointly by what pruning removes and what later adaptation restores.

Singh et al. [5] apply this pattern to AudioLDM by pruning U-Net channels and fine-tuning the reduced model on AudioCaps for 10^6 steps. Their released checkpoints recover strongly on the reported in-domain metrics. Our question begins after that result. Once recovery becomes part of the method, how stable is the *gain produced by recovery* when inference moves away from the setting at which the checkpoint is usually evaluated?

Fine-tuning gives a reason to expect variation. Adaptation can change pretrained representations differently across distributions [15], and text-to-audio systems are already sensitive to prompt and generation conditions. A post-recovery score therefore contains two kinds of information that should not be conflated. It describes the final checkpoint at one operating point, and it implicitly contains

the increment added by fine-tuning at that same point. Only the second quantity can tell us whether recovery generalizes across operating conditions.

2.2. Making recovery interpretable

No single automatic metric is sufficient for every aspect of generated audio. CLAP measures text-audio alignment [6]. FAD measures distributional distance and is sensitive to sample size and embedding choice [9–11]. Human-CLAP [13], PANNs [12] and FineLAP [14] provide complementary views of semantic alignment, event content and temporal grounding.

The important distinction is not between these metrics but between the quantities computed from them. The final score $S(P+FT)$ describes the checkpoint delivered to a user. The paired difference $S(P+FT) - S(P)$ describes what fine-tuning added at that operating point. A recovery ratio then asks how much of the gap from P to an interpretable reference was closed. We use the dense model, shuffled-caption chance and real audio as anchors so that a resolved numerical change can also be judged for scale.

3. EXPERIMENTAL DESIGN

3.1. Systems and operating points

The dense reference is AudioLDM-M-Full [4], with 415.96 M U-Net parameters. We study the two released pruning severities from Singh et al. [5]. Severity 1 retains 145.67 M parameters after 65.0% pruning, while severity 2 retains 71.08 M after 82.9% pruning. At each severity, P denotes the checkpoint before recovery and P+FT the released checkpoint after 10^6 AudioCaps fine-tuning steps. Their paired difference measures what the released recovery stage adds within the pruned architecture. It does not by itself identify an effect caused specifically by pruning.

The primary duration comparison uses 3.84 and 10.24 s. Severity 2 adds 5.12 and 7.68 s, producing a four-point sweep. AudioCaps TEST supplies the in-domain prompts [16]. Clotho provides a held-out sound-event domain with a different caption distribution [17]. A second held-out battery uses hip-hop and rap captions from MusicCaps [18]. The main severity 2 AudioCaps analysis uses 192 prompts, Clotho uses 96, and the extended hip-hop battery uses 127.

Generation uses DDIM [19] with 50 steps, guidance 2.5 and $\eta = 0$. A pre-specified robustness check repeats the severity 2 endpoint comparison with the published DDIM 200, guidance 3.5 recipe.

3.2. Interventions and follow-up checks

The mechanism and boundary checks were specified after the original operating-point result but frozen before their own generation and scoring. They therefore test explanations of the primary finding without being folded into its original confirmatory family.

The central intervention is symmetric. Starting from the same severity 2 P checkpoint, we run two full-U-Net fine-tunes for 20,000 steps with the same optimizer and data recipe. One uses random 3.84 s AudioCaps crops and the other uses 10.24 s examples. Both checkpoints are evaluated at both durations on the same 192 prompts with common generation noise. Let $J_{3.84}$ and $J_{10.24}$ denote their long-minus-short recovery interactions. Training-duration specialization predicts

$$\Delta J = J_{10.24} - J_{3.84} > 0, \quad (1)$$

because moving the training duration to 10.24 s should move recovery toward that evaluation duration.

A reduced dense 2×2 repeats the two 20,000-step fine-tunes from AudioLDM-M-Full. This experiment is diagnostic rather than a substitute for the unavailable dense checkpoint from the released 10^6 -step recovery. The dense baseline is EMA-averaged, whereas the short experimental exports are raw weights, and the dense model had already received substantial AudioCaps training. We therefore use this 2×2 to test duration specialization at the same short budget, not to estimate the million-step dense recovery that no longer exists.

Additional checks extend the released recovery to 15.36 s on 96 matched prompts, enlarge severity 1 with a disjoint 96-prompt sample, and complete dense hip-hop anchors for all 127 prompts. These checks address response shape, cross-severity stability and metric-floor interpretations.

3.3. Pairing, endpoints and estimands

Within each operating point, compared checkpoints receive identical diffusion noise x_T for a given prompt. The prompt is the statistical unit, and 95% intervals use a prompt-level percentile bootstrap with $B = 10^4$ resamples. This common-random-number design makes R a paired effect rather than a difference between independently sampled generation sets.

The primary endpoint is fused CLAP cosine from `laion/clap-htsat-fused`, revision 365dea6e [6]. Human-CLAP provides a second alignment scorer. KL is measured against matched real references, with positive recovery meaning that P+FT reduces KL relative to P. For PANNs, *top-10 capture* is the number of ground-truth AudioSet labels from the reference clip that appear among the generated clip’s ten highest-scoring PANNs classes. PANNs capture recovery is the paired increase in that count from P to P+FT.

For operating point o , recovery gain is

$$R(o) = \mathbb{E}_i [S(P+FT, i, o) - S(P, i, o)]. \quad (2)$$

The duration interaction is $J = R(10.24\text{ s}) - R(3.84\text{ s})$. We also report

$$\rho_{\text{ref}} = \frac{R}{S(\text{ref}) - S(P)}, \quad (3)$$

which expresses the gain as a fraction of the available gap to a dense or real-audio reference.

4. RESULTS

4.1. Recovery changes with requested duration

The released severity 2 recovery shows a large duration interaction. CLAP gain rises from $R = +0.085$ [$+0.066, +0.105$] at 3.84 s to $+0.244$ [$+0.215, +0.273$] at 10.24 s, giving $J = +0.159$ [$+0.131, +0.188$]. The intermediate points in Table 1 show that this is a gradual response rather than a difference created only by the two endpoints. The scale also changes materially. Recovery closes 44% of the gap from P to dense at 3.84 s and 82% at 10.24 s.

The same pattern appears outside the primary scorer. Human-CLAP, KL to matched real references and PANNs top-10 capture all show a larger recovery at longer durations. The published DDIM 200, guidance 3.5 recipe reproduces the endpoint interaction with $J = +0.184$ [$+0.126, +0.243$], and the registered severity 2 conclusions survive Holm correction. The final step of the secondary-metric sweep is not resolved, so these measures support duration de-

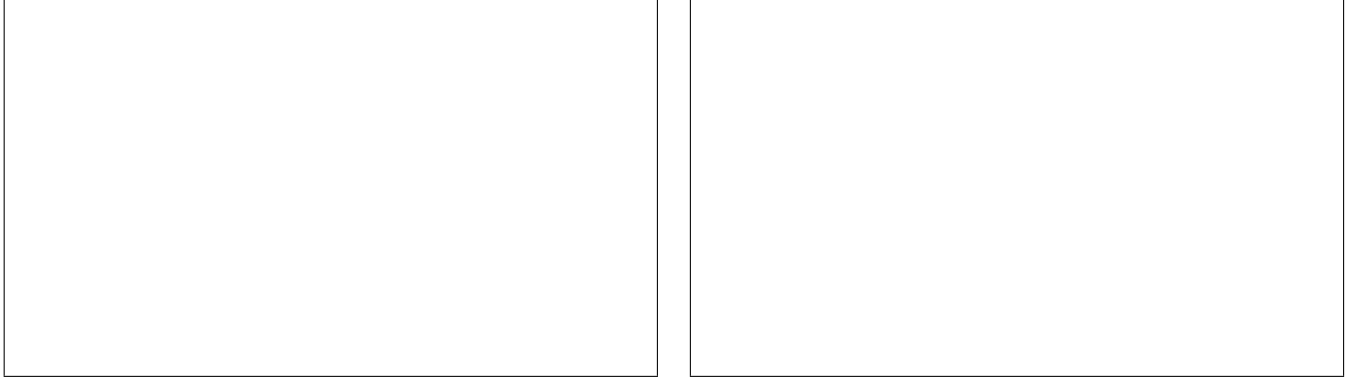


Fig. 1. Duration dependence and the symmetric training-duration test. Left, the released severity 2 recovery is evaluated across duration, with the 15.36 s extension compared on its matched 96-prompt subset. Right, two 20,000-step fine-tunes from the same pruned checkpoint differ only in training duration and are evaluated at both 3.84 and 10.24 s.

pendence without requiring every scorer to keep increasing through 10.24 s.

Extending generation beyond the native duration does not reveal a sharp peak. On the matched 96-prompt subset, recovery changes by $+0.021$ $[-0.023, +0.067]$ from 10.24 to 15.36 s. The interval is too wide to establish a plateau, but it provides no clear evidence of another increase comparable with the earlier sweep steps.

4.2. Training duration does not determine the favorable evaluation duration

The matched 20,000-step intervention supplies the direct test. The checkpoint trained at 3.84 s gains $+0.009$ $[-0.006, +0.024]$ when evaluated at 3.84 s and $+0.075$ $[+0.053, +0.097]$ at 10.24 s, giving $J_{3.84} = +0.065$ $[+0.044, +0.087]$. The checkpoint trained at 10.24 s also gains more at the longer evaluation duration, but its interaction is smaller, $J_{10.24} = +0.031$ $[+0.010, +0.052]$. Their paired difference is $\Delta J = -0.035$ $[-0.064, -0.004]$.

This comparison is the quantity that training-duration specialization predicts directly. Moving the training duration from 3.84 to 10.24 s should increase the long-minus-short interaction, which would give $\Delta J > 0$. The observed value is resolved in the opposite direction. At this matched 20,000-step budget, the duration effect therefore does not follow the duration used for fine-tuning.

The reduced dense 2×2 does not provide the missing causal control for pruning. Both 20,000-step dense fine-tunes strongly reduce CLAP relative to the EMA dense baseline, with gains around -0.2 . The training loss remains stable, which is consistent with the anticipated raw-weight-versus-EMA mismatch and limited AudioCaps headroom rather than optimizer divergence. Its only clean duration statement is negative. The dense specialization contrast is $\Delta J_{\text{dense}} = -0.005$ $[-0.034, +0.024]$, showing no resolved dependence on training duration at this short budget. The unavailable dense checkpoint after the released 10^6 -step recovery remains the relevant missing control.

4.3. Recovery transfers selectively across domains

Clotho shows that leaving AudioCaps does not by itself remove recovery. At 10.24 s, Clotho gives $R = +0.210$ $[+0.176, +0.243]$ and closes 74% of the dense gap. On the matched 96-prompt comparison, AudioCaps exceeds Clotho recovery by only $+0.032$ $[-0.023, +0.088]$, so the difference is unresolved.

Hip-hop presents a substantially weaker transfer case. With all 127 prompts, recovery is $+0.026$ $[+0.007, +0.044]$ at 3.84 s and $+0.027$ $[+0.003, +0.050]$ at 10.24 s. Dense alignment remains well above shuffled-caption chance at both durations, by about $+0.11$, so the small recovery cannot be dismissed as a floor-limited battery. At full coverage, recovery closes about 11% of the dense gap at 3.84 s and 12% at 10.24 s. The domain result is therefore a transfer profile rather than a binary in-domain versus out-of-domain effect.

4.4. Cross-severity and short-generation diagnostics

Severity 1 confirms that prompt selection matters to the estimated interaction. The original outcome-blind 80-prompt draw gives $J = +0.044$ $[-0.000, +0.088]$, while a disjoint 96-prompt draw with a different hash salt and a five-caption-row selection rule gives $+0.169$ $[+0.115, +0.222]$. Their unpaired difference is $+0.124$ $[+0.058, +0.194]$. The pooled $n = 176$ estimate, $+0.112$ $[+0.076, +0.149]$, is therefore useful as evidence that the interaction can resolve at severity 1, but not as evidence that its magnitude is stable across prompt samples.

The short-duration effect is not explained by a collapsed dense baseline under CLAP. The dense model’s floor-corrected response from 3.84 to 10.24 s is $+0.142$ $[+0.111, +0.172]$, close to the real-audio crop response of $+0.150$ $[+0.133, +0.166]$. A separate crop analysis nevertheless shows that recovery itself depends on how the audio is generated. Scoring the first 3.84 s of a 10.24 s generation gives $R_{\text{crop}} = +0.172$ $[+0.150, +0.194]$, which is $+0.087$ $[+0.065, +0.110]$ above recovery from a separately generated 3.84 s clip.

5. DISCUSSION AND LIMITATIONS

The most reproducible observation in this study is not a preferred training duration. It is that the gain added by fine-tuning depends strongly on the duration at which the model is evaluated. The symmetric intervention makes the distinction testable. Specialization predicts that moving the fine-tuning duration from 3.84 to 10.24 s should move recovery toward the longer evaluation point. Instead, both fine-tunes gain more at 10.24 s and the long-trained checkpoint has the smaller duration interaction.

These results motivate a narrower hypothesis about *expression of adaptation gains*. Fine-tuning may produce changes whose measurable benefit is amplified at some inference durations, even when those durations were not favored during adaptation. This account is

Table 1. Recovery across duration and domain. All intervals are 95% prompt-bootstrap intervals. Positive KL recovery means that P+FT reduces KL to the matched real reference relative to P. Human-CLAP, KL and PANNs sweep rescoring are corroborative analyses.

Duration	CLAP R	Human-CLAP R	KL recovery	PANNs capture R
3.84 s	.085 [.066, .105]	.189 [.162, .217]	.66 [.43, .92]	.19 [.06, .32]
5.12 s	.139 [.115, .164]	.278 [.243, .314]	1.16 [.91, 1.41]	.38 [.23, .53]
7.68 s	.201 [.175, .227]	.371 [.336, .405]	1.95 [1.68, 2.23]	.76 [.61, .91]
10.24 s	.244 [.215, .273]	.375 [.340, .409]	2.22 [1.92, 2.52]	.86 [.70, 1.02]
J	.159 [.131, .188]	.185 [.151, .220]	1.56 [1.20, 1.92]	.67 [.49, .85]
Domain	n	$R(3.84\text{ s})$	$R(10.24\text{ s})$	$\rho_{\text{dense}}(10.24\text{ s})$
AudioCaps	192	.085 [.066, .105]	.244 [.215, .273]	.82
Clotho	96	.098 [.072, .125]	.210 [.176, .243]	.74
Hip-hop	127	.026 [.007, .044]	.027 [.003, .050]	.119 [.015, .215]

consistent with the matched 20,000-step intervention and with the public dense text-fine-tuned reference, which also shows a larger gain at 10.24 s. It is not equivalent to saying that short generation is broken. Under the primary CLAP endpoint, the dense model’s own duration response is close to the real-audio response, so base-model degradation at 3.84 s does not explain the interaction. The mechanism that makes adaptation gains more visible at longer inference durations remains open.

The domain results illustrate why the same evaluation principle matters beyond duration. Clotho retains most of the AudioCaps recovery, whereas hip-hop retains only a small fraction of the dense gap despite a discriminating metric. A single label such as “out of domain” would hide this difference. Replications should therefore move along more than one domain axis and retain a dense or real-data anchor whenever a small gain could otherwise be mistaken for a metric floor.

The main causal limitation also remains clear. The experimental dense 2×2 uses a short raw-weight fine-tune against an EMA baseline and substantially degrades the dense model, so it cannot replace the unavailable dense checkpoint after the released 10^6 -step recovery. The matched 20,000-step pruned intervention is strong evidence against training-duration specialization at that budget, but it does not determine whether the million-step duration dependence is specific to pruning, to fine-tuning in general, or to another property of AudioLDM.

Other limits concern generality and power. The study covers one model family, two pruning severities and a small domain grid. The severity 1 interaction varies substantially across two outcome-blind prompt samples. The step from 10.24 to 15.36 s is too uncertain to establish a plateau. Human-CLAP, KL and PANNs corroborate the duration pattern but are still automatic metrics, while the small blinded author listening exercise is descriptive only.

The practical implication is simple. A recovery claim should report the paired increment from P to P+FT at the intended operating point and at deliberately displaced points. Common generation noise improves the comparison, and dense, chance or real-data anchors make the size of the gain interpretable. If a mechanism is proposed, its intervention should change a directional prediction rather than merely add another correlation. Here, matching the fine-tuning budget while changing training duration was the decisive test.

6. CONCLUSION

Recovery fine-tuning gain is strongly operating-point dependent in the AudioLDM-M checkpoints studied here. The duration interaction survives multiple scorers and both pruning severities, while a

symmetric intervention shows that it does not follow training duration at a matched 20,000-step budget. Recovery transfers strongly to Clotho and only weakly to hip-hop despite informative dense anchors. Reporting recovery as a paired function of operating point therefore gives a more faithful account of compressed-model behavior than a single post-fine-tuning score.

7. REFERENCES

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